🚀 COMPLETE AGENTIC AI ROADMAP (0 → Production Engineer)

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NOTE:- if this roadmap is shared across channels then don't forget to give the credits alongside it.

🟢 PHASE 0 — Python (Only What You Actually Need)

Duration: 1 week

1️⃣ Core Python

- Data types (list, dict — especially dict)
- Loops
- Functions
- List comprehensions
- JSON handling

2️⃣ OOP (Important)

- Classes
- Dataclasses
- Inheritance
- Composition

3️⃣ Async Programming

- async / await
- asyncio
- httpx (async HTTP)

4️⃣ APIs

- FastAPI basics
- POST endpoints
- Pydantic models

5️⃣ Packaging

- venv
- pip
- requirements.txt

You're done with Python when you can:

- Build a REST API
- Call an LLM API
- Handle errors properly

🟡 PHASE 1 — LLM & Prompt Foundations

Duration: 1 week

Learn deeply:

- Tokens & context window
- Temperature
- System vs user prompts
- Few-shot prompting
- Function calling
- Tool calling
- Structured output parsing

Then build:

- Basic chatbot
- Streaming response
- Structured JSON output generator

🔵 PHASE 2 — RAG Core (Data Injection Layer)

This matches what you asked earlier.

Duration: 1 weeks

📦1️⃣Document Loaders

Learn:

- PDF loaders

- Directory loader
- CSV / JSON loader
- Web loader
- S3 loader

Build:

- Chat with PDF
- Chat with folder of documents
- Web page ingestion

🌐2️⃣ Web Document Loading

Learn:

- requests + BeautifulSoup
- Playwright (for dynamic pages)
- Cleaning HTML
- Removing scripts/styles

✂️3️⃣ Text Splitting

Learn:

- RecursiveCharacterTextSplitter (most used)
- Token-based splitting

- Chunk overlap
- Semantic splitters

Understand:

- Why chunk size matters
- Why overlap improves retrieval

🧠4️⃣ Embeddings

Experiment with:

- HuggingFace (local)
- Ollama (local)
- OpenAI
- Gemini

Understand:

- Cosine similarity
- Vector dimension
- Tradeoff: cost vs latency vs quality

🗄️5️⃣ Vector DB

Start with:

- ChromaDB

- FAISS

Then understand:

- Metadata filtering
- Hybrid search
- Top-k retrieval

🏗️ Build These RAG Systems

1. Basic RAG
2. Conversational RAG
3. Multi-document RAG

Then move to:

- Adaptive RAG
- Corrective RAG
- Agentic RAG

(You already have these files — perfect.)

🟣 PHASE 3 — LangGraph Fundamentals

Now we move into real agent engineering.

Duration: 1.5 weeks

From your structure:

📊1️⃣ LangGraph Basics

- Nodes
- Edges
- State
- State schema
- Conditional edges
- Graph execution

Understand:

- State machine mindset
- Deterministic flows

Build:

- Simple state graph
- Graph with branching
- Graph visualization

🤖2️⃣ LangGraph Chatbot

Learn:

- Stateful conversations
- Message history

- Persistent memory
- Message trimming

Build:

- Chatbot with memory
- Session-based conversation

🧩3️⃣ LangGraph Components

From your folder:

- Tools with LLM
- Multiple tools
- ReAct agent
- Agent with memory
- Streaming in LangGraph

Understand:

- Tool binding
- Tool execution loop
- ReAct reasoning
- Tool call JSON schema

Build:

- Calculator tool

- Web search tool
- Database tool

Then:

- Multi-tool agent

🟠 PHASE 4 — Workflows

Now you stop building “chatbots”
You start building “systems”.

Duration: 1.5 weeks

From your folder:

🔹 Prompt Chaining

- Sequential steps
- Output of one → input of next

🔹 Routers

- Conditional branching
- Classifier → route to different chains

🔹 Parallelization

- Run multiple tasks in parallel
- Combine results

Understand:

- Latency optimization
- Deterministic pipelines
- Fan-out / fan-in pattern

Build:

- Document analyzer workflow
- Multi-step research agent

🔴 PHASE 5 — Orchestrators

Now we go multi-agent.

Duration: 1 weeks

From your structure:

- Orchestrator → Worker pattern

Learn:

- Task decomposition
- Delegation
- Planner → Executor
- Multi-agent coordination

Build:

- Research agent
- Blog generator agent (planner + writer + editor)

Understand:

- When to orchestrate
- When not to over-engineer

🟡 PHASE 6 — Evaluator & Optimizer

This is where 90% people stop.
You must NOT.

Duration: 1 weeks

Learn:

- LLM as judge
- Grading outputs
- Binary scoring
- Confidence scoring
- Self-reflection loop

Build:

- Auto-evaluation pipeline
- Retry on low score
- Self-improving RAG

Understand:

- Hallucination detection
- Retrieval grading

- Response grading

This is production-level thinking.

🟢 PHASE 7 — Human in the Loop (HITL)

Very important for enterprise AI.

Learn:

- Manual approval nodes
- Interrupt and resume
- State checkpointing
- Editable outputs

Build:

- Agent that pauses before final response
- Admin review system

🔵 PHASE 8 — Advanced RAG (From Your 9-RAG Folder)

Now combine everything.

Learn:

1️⃣ Adaptive RAG

- Decide whether to retrieve or not

2️⃣ Corrective RAG

- Evaluate retrieval quality
- Retry retrieval if weak

3️⃣ Agentic RAG

- Agent decides:
    - Which tool?
    - Which retriever?
    - Which source?

Build:

- Multi-source RAG (PDF + Web + DB)

🟣 PHASE 9 — Debugging & Observability

From your Debugging folder:

Learn:

- LangGraph visualization
- Tracing
- Logging
- State inspection
- Token tracking

Understand:

- Why agent debugging is hard
- How to isolate failures

⚫ PHASE 10 — Production Agent Engineering

Now think like system designer.

Learn:

- Rate limiting
- Caching embeddings
- Cost optimization
- Retry strategies
- Exponential backoff
- Guardrails
- Output validation
- Prompt injection protection

Build:

- Production-ready RAG API
- Agent service with monitoring

🧠 Phase 11 — Scaling & Architecture Thinking

Since you like system design:

Learn:

- Agent memory persistence
- Redis for session state
- Background workers
- Queue systems
- Distributed vector DB
- Horizontal scaling

Design:

- Instagram-like AI assistant architecture
- Enterprise knowledge base system

🎯 Final Skill Levels

If you complete all phases:

You can build:

- Enterprise RAG system
- Multi-agent orchestration
- Self-correcting AI workflows

- HITL review systems
- Production-grade AI backend

You'll not just "use LangChain".

You'll engineer AI systems.

⏳ Realistic Timeline

If consistent:

- Python → 1 weeks
- LLM + RAG → 1 weeks
- LangGraph → 1.5 weeks
- Workflows + Orchestrators → 1.5 weeks
- Eval + HITL → 1.5 weeks
- Production → ongoing

👉 1-2 months to become strong

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RESOURCE :- https://share.google/jg2EStPLpAVOvJuo1